

BANK LOAN DEFAULT RISK ANALYSIS

End-to-End Data Analytics & Business Intelligence Project

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USN	23BBTCS124
Program	B.Tech — Computer Science & Engineering
Tools	Python • Pandas • Matplotlib • Power BI • DAX
Dataset	255,347 loan records

Project Focus

Identify borrower and loan characteristics associated with higher observed default rates, segment portfolio risk, and translate the analysis into practical risk-management actions.

Project Report • September 2026

1. Executive Summary

This project analyzes historical loan records to understand patterns associated with loan default. It was designed as a business analytics problem rather than only a prediction exercise. Python and Pandas were used for data inspection, cleaning checks, feature engineering and exploratory analysis, while Power BI and DAX were used to communicate the results through an interactive four-page report.

The dataset contains **255,347 borrowers** and **29,653 defaults**, producing an overall observed default rate of **11.61%**. A deeper segmentation identified a group defined by **Low Income + High Loan Amount + Highest 25% Interest Rate**. This segment contains **3,972 borrowers** and **1,460 defaults**, with an observed default rate of **36.76%**.

The segment's rate is approximately **3.17×** the overall observed rate, but it contributes only **4.92%** of all defaults. Therefore, the project does not treat one segment as a complete explanation of portfolio risk. Instead, it supports a broader multi-factor risk-management approach.

255,347	29,653	11.61%	36.76%
Borrowers	Defaults	Overall Rate	Segment Rate

2. Business Problem

Financial institutions need to understand which characteristics are associated with repayment risk so that applications and existing loans can receive appropriate assessment, structuring and monitoring.

The project addresses these business questions:

- What is the overall distribution of loan defaults?
- How does the observed default rate vary across borrower and loan characteristics?
- Which combinations of characteristics are associated with substantially higher observed default rates?
- Does one high-risk segment account for most defaults, or are defaults distributed more broadly?
- How can the findings support a practical risk-management strategy?

Analytical principle: the analysis identifies associations in historical data. A higher observed default rate does not by itself prove that a characteristic causes default.

3. Project Objectives

- Understand the structure and quality of the loan dataset.
- Validate the target variable and quantify overall default levels.
- Perform exploratory analysis across numerical and categorical variables.
- Create interpretable groups for important numerical variables.
- Identify a high-risk combination using observed default-rate analysis.
- Examine defaults outside the defined high-risk segment.
- Translate analytical findings into practical business recommendations.
- Build an interactive Power BI report for decision support.

4. Dataset Description

The project notebook contains 255,347 records and 18 original columns covering borrower characteristics, loan characteristics, financial indicators, supporting information and the default outcome.

Category	Variables
Identifier	LoanID
Borrower / financial profile	Age, Income, CreditScore, MonthsEmployed, NumCreditLines
Loan characteristics	LoanAmount, InterestRate, LoanTerm, LoanPurpose
Financial burden	DTIRatio
Demographic / employment	Education, EmploymentType, MaritalStatus
Existing obligations	HasMortgage, HasDependents
Support	HasCoSigner
Target	Default

The notebook's structural inspection reports 255,347 non-null values for each of the 18 original columns, with 8 integer columns, 2 floating-point columns and 8 object/string columns.

5. Data Understanding & Data Quality

The analysis followed standard checks including shape, column names, data types, descriptive statistics, unique values and target distribution before deeper analysis.

Check	Project result / treatment
Dataset size	255,347 rows × 18 original columns
Missing values	No missing values reported across the original 18 columns
Duplicate rows	No duplicate rows in the completed quality check
Target	Default; 0 = non-default and 1 = default
Target count	29,653 defaults
Overall default rate	11.61%
Derived features	Grouped variables for age, income, loan amount, credit score, DTI, interest rate and loan term

Grouped variables were created to make numerical patterns easier to interpret in charts and business discussions. They are analytical representations of the original variables, not replacements for them.

6. Exploratory Data Analysis

The analysis examined observed default rates across major borrower and loan dimensions. The main patterns were:

Variable	Lower / relevant group	Higher / relevant group
Age	Highest 25%: 5.36%	Lowest 25%: 19.15%
Income	High: 9.02%	Low: 17.38%
Loan amount	Low: 8.25%	High: 15.60%
Interest rate	Lowest 25%: 6.58%	Highest 25%: 17.52%
Credit score	800–899: 9.81%	300–399: 13.34%

Interpretation: several variables show meaningful differences in observed default rates across their defined groups. These patterns are useful for segmentation and investigation, but they do not establish causality.

7. Risk Segmentation

The project moved beyond one-variable analysis by combining three grouped variables. The selected segment was defined as:

Low Income + High Loan Amount + Highest 25% Interest Rate

Metric	Value
Borrowers in segment	3,972
Defaults in segment	1,460
Observed default rate	36.76%
Overall observed default rate	11.61%
Relative rate	3.17× overall rate
Share of all defaults	4.92%

Metric	Value
Defaults outside segment	28,193 (95.08%)

Within the defined segment, observed default rates across DTI groups were 34.88%, 36.96%, 37.06% and 38.16%. The elevated rate is therefore present across all four DTI groups rather than being isolated to one group.

8. Portfolio-Wide Analysis

After isolating the defined high-risk segment, 28,193 defaults remain outside it, representing 95.08% of all defaults. The remaining-default analysis shows that defaults are distributed across multiple profiles.

Dimension	Largest concentration among remaining defaults
Age group	Lowest 25% — 12,575 (44.60%)
Income group	Low — 9,635 (34.18%)
Loan amount group	High — 8,501 (30.15%)
Interest rate group	Highest 25% — 9,696 (34.39%)
DTI group	Highest 25% — 7,307 (25.92%)
Employment type	Unemployed — 8,205 (29.10%)
Marital status	Divorced — 10,156 (36.02%)
Loan purpose	Business — 6,029 (21.38%)

9. Key Findings

- The overall observed default rate is 11.61% across 255,347 records.
- The defined high-risk segment has an observed default rate of 36.76%, compared with 11.61% overall.
- The segment contains 3,972 borrowers and 1,460 defaults.
- The segment contributes only 4.92% of all defaults, so it cannot explain the majority of portfolio defaults.
- Observed default rates vary substantially across several grouped numerical variables.
- The remaining-default analysis supports a multi-factor risk-management approach rather than a single-rule approach.

10. Business Recommendations

The analysis can be translated into a staged risk-management process:

Stage	Recommended action
Identify	Flag combinations associated with materially higher observed default rates for additional review.
Assess	Review repayment capacity, existing obligations, loan amount relative to income, employment stability, DTI and credit history.
Structure	Where appropriate, consider loan amount, pricing, term or other product-structure decisions based on verified risk assessment.
Monitor	Use stronger post-approval monitoring for portfolios or profiles showing elevated historical risk.
Validate	Recalculate the patterns periodically using newer data and test whether the observed relationships remain stable.

Recommendation boundary: the analysis should support enhanced assessment and monitoring, not automatic rejection based on a single segment label. Any operational policy should be validated against current data, business rules, legal requirements and appropriate fairness controls.

11. Power BI Decision-Support Report

The final Power BI report was organized into four pages so the audience can move from portfolio overview to risk concentration, broader portfolio analysis and business action.

Page	Purpose	Main content
1 — Executive Overview	Portfolio summary	5 KPI cards and default-rate analysis across major grouped variables
2 — Risk Analysis	High-risk segment deep dive	Segment KPIs plus Employment Type, DTI Group, Loan Purpose and Education
3 — Portfolio-Wide Analysis	Defaults outside defined segment	Remaining-default counts across Age, Income, Loan Amount, Interest Rate, Employment and DTI
4 — Business Solution	Translate analysis into action	Key finding, risk strategy, portfolio limitation and expected business impact

Core DAX measures include Total Borrowers, Total Defaults, Default Rate, Default Rate Chart, Average Loan Amount and Average Interest Rate. A calculated Risk Segment column distinguishes the defined high-risk segment from the rest of the portfolio.

12. Limitations

- The analysis is based on historical records and describes observed relationships; it does not establish causation.
- Grouped variables simplify continuous measurements and can hide variation within each group.
- The defined high-risk segment is an analytical segment, not a production underwriting rule.
- The dataset and analysis do not by themselves establish how stable these relationships would be on future portfolios.
- A production risk system would require validation, monitoring, governance, fairness assessment and business-policy controls.

13. Conclusion

The Bank Loan Default Risk Analysis project demonstrates an end-to-end analytics workflow connecting data quality, exploratory analysis, segmentation, business interpretation and dashboard communication. The analysis identified a specific combination of low income, high loan amount and highest-quartile interest rate with a substantially higher observed default rate than the portfolio average.

At the same time, the majority of defaults occur outside this segment. The final business conclusion is therefore to use the identified segment as an enhanced-review signal within a broader risk-management framework: **identify** → **assess** → **structure appropriately** → **monitor** → **validate**.

14. Appendix — Derived Analytical Groups

Derived variable	Categories / method
CreditScoreGroup	300–399, 400–499, 500–599, 600–699, 700–799, 800–899
IncomeGroup	Quartiles: Low, Lower-Middle, Upper-Middle, High
LoanAmountGroup	Quartiles: Low, Lower-Middle, Upper-Middle, High
DTIGroup	Quartiles: Lowest 25%, 25–50%, 50–75%, Highest 25%
InterestRateGroup	Quartiles: Lowest 25%, 25–50%, 50–75%, Highest 25%
AgeGroup	Quartiles: Lowest 25%, 25–50%, 50–75%, Highest 25%
LoanTermGroup	Quartiles: Lowest 25%, 25–50%, 50–75%, Highest 25%

Source: Bank Loan Data.ipynb - Colab.html and the completed Power BI analysis. Numerical findings in this report are based on the completed project analysis.